

Mathematical Foundations: Status-Labeled Definitions, Propositions, Conjectures, and Open Problems

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Abstract

States Fractal Resonance Coherence as a mathematical research program with explicit status labels. Every claim is a Definition, Proposition, Conjecture, or Open Problem. Four propositions are recorded: a two-pole interior band, a forced cubic term in non-even coherence expansions, critical slowing near coherence transitions, and the von Mises exchange-rate identity $dS/d \ln C = -\kappa r$ with a single closure point at $\kappa = 1.608279$, $C = 0.621782$ for $k^ = 1$. Two negative computational results are recorded at equal volume: $\Lambda = \Lambda_0 \ln C$ is not a dual-affine coordinate and not the Fisher potential on the von Mises family. The half-line obstruction for a fundamental*

scalar Lambda field is quantified, and six precise open problems are left where the program can be advanced or killed.

1. Intent

This paper states the mathematical spine of Fractal Resonance Coherence. It is not a proof that the physical conjectures are true. It is a status-labeled map of the program.

Every claim is assigned one of four statuses:

Status	Meaning
Definition	a convention or mathematical object introduced by the framework
Proposition	a standard computation, symmetry argument, or directly checked identity
Conjecture	a physical claim not yet proven
Open problem	a precise task left for proof, disproof, or refinement

The discipline matters. A foundations paper should not hide conjectures inside theorem language. The aim is that a mathematically trained reader can see exactly where to push.

2. Coherence and the Lambda Field

For phases $\phi_1, \dots, \phi_N$ on the circle, a basic coherence parameter is the Kuramoto order-parameter magnitude

$$C = \left| \frac{1}{N} \sum_{j=1}^N e^{i\phi_j} \right|, \quad C \in (0,1].$$

The scalar coherence field is

$$\Lambda = \Lambda_0 \ln C, \quad \Lambda_0 > 0,$$

so $\Lambda \leq 0$ when $C \leq 1$: full coherence is $\Lambda = 0$, the incoherent limit is $\Lambda \rightarrow -\infty$. The constant k^* is a **units bridge** — it makes entropy and $\ln C$ additively comparable. It is not a fitted coupling.

3. The Open Entropy–Coherence Ledger

The naive expression $dS + k^d \ln C = d(S + k^d \ln C)$ is exact if S and C are state functions; treating that exact differential as a physical residual would be circular. The FRC ledger is **directional**, not a closed identity. The secure content is the second-law inequality

$$\sigma_{ST} \equiv dS_{\text{total}} = d_i S \geq 0,$$

and the **combined ledger current** $\sigma_{566} = \sigma_{ST} + k^d \ln C$ is not sign-definite in general — its sign is directional: $\sigma_{566} \geq \sigma_{ST} \geq 0$ when coherence is gained, and $\sigma_{566} \rightarrow k^d \ln C < 0$ when coherence is destroyed quasi-statically. The residual is a **measured** current, not an identity: writing $\lambda = -d_e S$ (so $d_i S + k^d \ln C = \lambda$) forces the conserved closed form $dS + k^d \ln C = 0$, which the open-system reading and direct measurement both reject; it is admissible only as the reversible boundary case. The stronger claim — that the combined current obeys a law of nature with a single units bridge k^* — is a **conjecture**, not a theorem, and its erasure-side (scale-free floor) version was instrumented and **falsified** (100.005 v2.1), leaving the directional/structural conjecture as what survives.

4. Effective Dynamics

At the effective level, the Λ field is modeled by a relaxation-diffusion-advection closure:

$$\tau_{\Lambda} (\partial_t \Lambda + \mathbf{u} \cdot \nabla \Lambda) = D_{\Lambda} \nabla^2 \Lambda - \Lambda + \Lambda_{\text{eq}}[\rho, \psi, E] + \xi_{\Lambda}.$$

This places the effective dynamics in the reaction-diffusion / time-dependent Landau–Ginzburg family. A proposed fundamental scalar Lagrangian

$$\mathcal{L}_{\Lambda} = \frac{1}{2} (\partial_{\mu} \Lambda) (\partial^{\mu} \Lambda) - V(\Lambda), \quad V(\Lambda) = \frac{1}{2} m_{\Lambda}^2 \Lambda^2 + \frac{\lambda_4}{4!} \Lambda^4$$

carries a known tension: the field theory ranges over real Λ , while the coherence definition requires $\Lambda \geq 0$ (Section 8).

5. The Four Propositions

Proposition 1 — Two-pole interior band. Let $F(C)$ be a bounded functional measuring integration that preserves differentiation, $C \in (0, 1)$, with both endpoints vanishing: $F(0^+) = 0$, $F(1^-) = 0$. Then F attains an interior maximum on $(0, 1)$. The established statement is the existence of an interior band, not a universal numerical coordinate for it — in particular, no golden-ratio value is asserted.

Proposition 2 — Forced cubic. For a coherence expansion $C(m)=1+c_1m+c_2m^2+c_3m^3+O(m^4)$ and reciprocity entropy term $U(m)=-k^*\ln C(m)$, the cubic coefficient is

$$a_3=\frac{k^*}{3}(-c_1^3+3c_1c_2-3c_3).$$

Unless the odd part of $\ln C$ vanishes through third order, an odd cubic term is forced. The deterministic check with $k^*=1$, $c_1=0.3$, $c_2=0.2$, $c_3=-0.1$ gives $a_3=0.151$. Physical consequences (first-order or tricritical behavior) must still be established per platform.

Proposition 3 — Critical slowing. Linearizing the effective dynamics around a fixed point Λ^* , if μ_1 is the leading eigenvalue, then near a coherence transition

$$\mu_1\rightarrow 0^-, \quad \tau_{\rm relax}\sim \frac{1}{|\mu_1|}\rightarrow\infty.$$

This is standard critical slowing; the FRC wager is that coherence transitions in selected systems show measurable slowing in the Λ channel before collapse or state change.

Proposition 4 — Von Mises exchange-rate identity. For the von Mises family $p(\phi)=e^{\{kappa\cos(\phi-\mu)\}}/(2\pi I_0(kappa))$ with coherence $r=C=I_1(kappa)/I_0(kappa)$ and entropy $S(kappa)=\ln(2\pi I_0(kappa))-kappa r$,

$$\{dS\over d\ln C\}=-kappa r,$$

so the ledger residual along the family is $\lambda=(k^*-kappa r)\,d\ln C$. For $k^*=1$, closure occurs at exactly one point:

Quantity	Value
$kappa$	1.608279
$C=r$	0.621782
$kappa r$	1.000000

This is the analytic form of the peak used in [[FRC-566-030]]. Closure is a special point, not a generic curve — supporting the open-system reading.

6. Negative Results: Lambda Is Not Canonical in Two Named Senses

The von Mises family is an exponential family: natural coordinate $kappa$, expectation coordinate $r=C$, Fisher potential $\ln I_0(kappa)$. Two candidate claims **fail** on this family:

Candidate claim	Result
$\ln C$ is a dual-affine coordinate	negative ; $r=C$ is the mixture-affine coordinate
$\ln C$ is the Fisher potential	negative ; $\ln I_0(\kappa)$ is the potential whose Hessian gives the Fisher metric

This is an important loss. It removes a tempting route by which $\Lambda=\Lambda_0 \ln C$ could be called canonical. The surviving information-geometric question is narrower: whether the open production form $d_i S + k^d \ln C$ has a connection, curvature, or holonomy interpretation on a suitable statistical bundle. These losses narrow the program rather than weaken it, and they are recorded at the same volume as the confirmations.

7. Conjectures

Conjecture	Current status
The open ledger holds as a law of nature with one units bridge k^*	useful and testable, but not proven; the von Mises rate κr is state-dependent and therefore a headwind to the strongest reading
Collapse dynamics recovers the Born rule as an attractor	supported only in equilibrium-limit arguments; a rigorous H-theorem remains open
Specific constants sit at coherence-ease thresholds	high risk; only defensible where the threshold is independently defined

Their failure would not erase the definitions or the propositions.

8. Half-Line Obstruction

The coherence definition requires $\Lambda \leq 0$, but a fluctuating real scalar field does not automatically respect that half-line. The deterministic appendix computes the positive-tail probability for an unconstrained Gaussian fluctuation around a negative mean:

Mean Λ	Fluctuation scale	Fraction with $\Lambda > 0$
-0.99	0.22	0.000003
-0.51	0.42	0.112319
-0.57	0.69	0.204377
-0.34	0.57	0.275424

When the fluctuation scale becomes comparable to the distance from the boundary, the field spends order-ten-percent time outside the physical region. Two paths exist: derive the half-line as an effective coarse-grained constraint, or reparametrize the fundamental field so $\Lambda \ll \ell_0$ intrinsically — at the cost of the clean polynomial scalar theory.

9. The Six Open Problems

Open problem	Task
UCC well-posedness	prove existence, uniqueness, regularity, and positivity for the effective Λ PDE
Born H-theorem	prove or disprove a Lyapunov functional whose equilibrium is the Born distribution
Connection or holonomy form	determine whether the open ledger is a connection or curvature object
Fundamental-to-effective coarse-graining	derive the effective UCC from a fundamental field while respecting $\Lambda \ll \ell_0$
Forced-cubic universality class	classify systems where the forced cubic changes transition order
Two-pole bifurcation normal form	classify how the interior maximum appears and moves with scale and coupling

These are the points where the mathematical program can be advanced or killed.

10. Reproducibility

The release package includes `pilot/foundations_checks.py`, which reproduces the forced-cubic coefficient, the von Mises exchange-rate identity by finite-difference check, the closure point, and the half-line obstruction fractions:

Check	Value
forced-cubic a_3 example	0.151000
exchange identity median relative error	4.57e-7
exchange identity max relative error, trimmed	1.41e-6
closure κ for $k^*=1$	1.608279
closure C for $k^*=1$	0.621782

11. Conclusion

The mathematical content of FRC, stated carefully, is not a finished theory. It is a structured research program: define coherence and the Lambda field; use the internal/external entropy split to avoid a circular closed ledger; place effective Lambda dynamics in known reaction-diffusion families; retain the established structural results; record the negative results at the same volume as the confirmations; and leave the physical claims as conjectures until the open problems are solved or killed.

Connections

- [[FRC-566-030]] — The von Mises/Kuramoto synchronization family whose peak Proposition 4 renders in analytic form
- [[FRC-566-001]] — The founding entropy–coherence reciprocity statement, here corrected into an open ledger
- [[FRC-566-010]] — UCC well-posedness, the first open problem's PDE line
- [[FRC-100-006]] — Born rule as resonant equilibrium; the Born H-theorem remains open
- [[FRC-100-007]] — Lambda-field formalism whose fundamental branch meets the half-line obstruction
- [[FRC-100-009]] — Local reciprocity dynamics using the effective Lambda closure

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